



Who lets ethics guide his economic decision-making? An empirical analysis of individual investments in ethical funds

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ABSTRACT

This paper shows that one out of eight investors chooses ethical funds and foregoes return for ethical principles. However, few screen completely. Human capital, being female and “empathetic” professions all predict ethical decision-making. Interestingly, neither income nor wealth explains screening.

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1. Introduction

Recent economics literature has directed much attention towards the way altruistic norms affect economic activity. Many individuals engage in altruistic activities such as contributing to charities (Andreoni, 2008), donating blood (Titmuss, 1970) or supplying volunteer labor (Freeman, 1997). A related phenomenon is preferences over ethics, where individuals integrate personal values and societal concerns with economic decision-making. For example, individuals may not want to partake in activities considered unethical or that have unwanted effects.

The habit of ethically restricting investments, generally referred to as *socially responsible investing*, elucidates ethical economic decision-making. This is an ancient phenomenon that has recently gained increased interest with more than 11% of all US funds managed with ethical guidelines in 2007 (Social Investment Forum, 2007). Given the rising importance of moral obligations to investors, effects on stock markets and banking as well as on company and government behavior may well arise.

Modern portfolio theory holds that, in general, diversification reduces risk and maximizes long-term returns. Consequently, anything that limits an investor's ability to diversify will unnecessarily

increase risk and yield a non-optimal portfolio. Hence, as integrating ethics with economic decision-making may incur a cost, it is a puzzle as to who actually let ethics govern their behavior.

This paper aims to empirically distinguish individual features predicting who allows ethics to guide their economic decision-making. This has been largely unexplored in economics. Yet, as the analysis is based on conjectures of what characterizes ethical investors, and is not a test of causality, the results should be seen as mainly descriptive.

2. Data

The data contain individual portfolio choices following a unique Swedish public-pension initiative launched in 2000, which allowed nearly the whole Swedish workforce to self-direct a part of their public-pension contributions.¹ Contributions could be invested in at most 5 funds out of over 600. Investors could then choose to screen their investments either to promote ethical behavior or to ban unethical behavior. Twenty funds had pronounced ethical investment profiles signified by name and by screen profile.² As the ethical

¹ Investors born between 1937 and 1954 had for each year a one percentage point lower investment share.

² Note that *Premievalsfonden* only has indirect ethical guidelines. As it is a replica of the default fund (constructed to reflect a national representative investor), it has no explicit ethical profile. All results are robust to its exclusion.

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Table 1
Summary statistics.

	Mean
Percentage with $\theta_{screen} > 0$	12.8
θ_{screen}	0.31 (0.21)
Observations	1387

Numbers in parentheses are standard deviations.

profiles were explicit, the probability of mistakenly choosing ethical funds was low. Those who made no active choice obtained a default alternative.

These data are merged with cross-sectional data from the Swedish Household Survey on Income for a representative sample in the year

1999. This consisted of around 38,000 individuals, of whom 46% were eligible, and of whom 62% chose to self-direct.

Before deciding to apply an ethical screen, the investor has to first decide whether to actively self-direct his contributions. $\Pr(\text{Active})_i$ is modeled as:

$$\Pr(\text{Active})_i = F_1(y(\text{fin. wealth}, \text{inc}, \text{edu}, \text{age}, \text{fem}), z(\text{pension}), T, e_i) \quad (1)$$

where i = non-participation.

This decision is based on familiarity with financial markets and pension-savings decisions, assuming that higher familiarity lowers transaction costs of self-directing. The proxies include financial wealth, a no financial wealth dummy, income, education, age, gender and tax-deductible pension-savings. As the policy was implemented

Table 2
Heckman model estimates on the probability of screening and on the share of ethical funds.

	Mean	Mean	Model I			Model II		
			$\Pr(\text{screen})_i$	$\Pr(\text{active})_i$	ME	θ_{screen}	$\Pr(\text{active})_i$	ME
Education 10–11 years	30%	34%	0.065 (0.053)	0.192 (0.029)***	5.2%	0.063 (0.021)***	0.181 (0.042)***	1.9%
Education 12 years	17%	18%	0.153 (0.059)**	0.154 (0.032)***	6.3%	0.083 (0.023)***	0.223 (0.049)***	1.6%
Education 15 years	21%	16%	0.265 (0.059)***	0.125 (0.034)***	8.3%	0.144 (0.022)***	0.356 (0.048)***	2.2%
Education > 15 years	20%	14%	0.355 (0.062)***	0.018 (0.038)	8.5%	0.159 (0.024)***	0.441 (0.052)***	2.0%
Female	60%	52%	0.145 (0.036)***	0.069 (0.021)***	15%	0.067 (0.014)***	0.181 (0.029)***	3.4%
Financial services	1%	3%	−0.364 (0.117)***		−1.4%	−0.008 (0.022)		
Manufacturing	20%	24%	0.016 (0.045)			0.002 (0.008)		
Health	34%	27%	0.106 (0.044)**		5.1%	0.030 (0.008)***		0.8%
Teaching	3%	2%	0.214 (0.097)**		0.9%	0.007 (0.016)		
Public administration	5%	4%	0.035 (0.080)			0.004 (0.014)		
Union	60%	56%	0.116 (0.056)**		2.8%	0.031 (0.010)***		0.4%
Income	223 389	217 769	0.005 (0.015)	0.152 (0.011)***	8.8%	0.015 (0.006)***	0.039 (0.012)***	3.0%
Net wealth	449 435	443 598	−0.001 (0.002)			0.001 (0.001)		
Age	40.6	40.7	−0.016 (0.015)	0.098 (0.006)***	21 %	0.021 (0.005)***	0.054 (0.009)***	8.3%
Age ²			−0.000 (0.000)	−0.001 (0.000)***	−12 %	−0.000*** (0.000)	−0.001 (0.000)***	−4.2%
Child	72%	71%	0.049 (0.042)			−0.018 (0.007)		
Financial wealth	215 947	235 103		0.004 (0.003)			0.002 (0.003)	
Financial wealth ²				−0.000 (0.000)			−0.000 (0.000)	
No financial wealth	27%	31%		−0.289 (0.022)***			−0.027 (0.014)*	
Pension-savings	52%	46%		0.354 (0.023)***			0.046 (0.014)***	
Time FE				Y			Y	
Constant			−1.543 (0.361)***	−1.699 (0.120)***		−1.076 (0.092)***	−2.808 (0.178)	
Artrho			−0.206 (0.108)*	−0.206 (0.108)***		3.141 (0.078)***	3.141 (0.078)***	
In sigma						−0.834 (0.026)***	−0.834 (0.026)***	
Obs	1387	9453	17,595	17,595		17,595	17,595	

Numbers in parentheses are standard errors. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. Columns 1 and 2 display mean values for active investors with or without a screen. Columns 3 and 4 display results from a Heckman selection probit-model on the probability to screen. Columns 6 and 7 display results from a Heckman selection model on the share of ethical funds. ME are statistically significant (5% level) differences in marginal effects between investors with or without a screen from a percentage change in each variable. Annual income (earnings, transfers), financial wealth (savings, bonds, funds/stocks), net wealth (financial wealth, real estate minus debt) are in 100,000 SEK. No financial wealth indicates a wealth below 3000 SEK. Pension-savings indicate having tax-deductible pension-savings.

in phases, time dummies T are included. e_i summarizes unobservable random attributes of self-directing investments.

Second, the investor chooses whether to screen, $\Pr(\text{screen})_j$ and, if so, by how large a share, θ_{screen} . The predictive characteristics are discussed below according to the following model:

$$\Pr(\text{screen})_j = F_2(i, \text{edu}, \text{union}, \text{empathetic} - \text{prof}', \text{fem}, \text{inc}, \text{wealth}, \text{age}, \text{child}, e_j) \quad (2)$$

where j = not choosing an ethical screen.

e_j summarizes the unobservable random attributes of choosing a screen.

3. Results

Table 1 reveals that one out of eight investors chooses ethical screening for about a third of the portfolio. Interestingly, this result is not driven by tax-related issues, and is based upon a large representative sample. Only 5% choose to screen completely, and hence, for the great majority, the ethical principle means having a fraction of one's portfolio invested ethically.

Following the two-stage decision process, a Heckman selection model is employed, with results displayed in Table 2. $\Pr(\text{Active})_i$ is estimated using a probit model, while $\Pr(\text{screen})_j$ and θ_{screen} are estimated using a probit and an OLS model. The Heckman modeling also demonstrates that active investors are typically more financially sophisticated and more actively concerned about their pensions.

Education is by far the most important predictor of applying ethical screening. Another important predictor is being female. Working in empathetic professions, such as health-care and teaching, also signifies investors who screen.³ These investors are then compared to investors who work in the service sector, and who thus similarly work with a focus on serving others, but with an ultimate profit-earning goal.

Focusing only on investors in the non-private sector, see Table 3, where the majority of empathetic professions are found, it is shown that providing indispensable public goods, such as national safety or rescue services, or working in religious associations, thus investors guided by ethical beliefs in their work-environment, clearly predict the choice to screen. Finally, having actively chosen to support a group-norm compared to an individual norm, here signified by trade union enrollment, is positively associated with the choice to screen.

Interestingly, neither income nor wealth affects the choice to screen. Yet, conditional on screening, investors with more income screen a larger share. Age is also positively associated with the proportion screened but this association declines for older investors. The latter is noteworthy because these investors are, by design of the public-pension initiative, allowed a smaller public-pension fraction to invest and thus have a lower cost of screening.⁴ Finally no correlation between screening and having children is present.

4. Conclusion

Many self-directing investors show a willingness to forego economic returns for non-economic values. This is noteworthy. Alike charitable giving, the choice to screen is a contribution to a public good, as the benefit of an ethics screen cannot exclude others. Consequently, there must be a private benefit of screening.

Table 3

Heckman model estimates on the probability of screening and on the share of ethical funds among non-private sector workers.

	Mean	Mean	Model I $\Pr(\text{screen})_i$	Model II θ_{screen}
State-owned corporation	2%	1%	0.389 (0.199)*	0.003 (0.039)
Local government	54%	54%	0.169 (0.065)**	0.031 (0.012)***
County council	20%	16%	0.241 (0.078)***	0.031 (0.013)**
Religious association	3%	1%	0.613 (0.168)***	0.020 (0.026)
Non-religious association	4%	5%	0.149 (0.125)	0.023 (0.023)
Heckman model			Yes	Yes
Obs	661	3621	6529	6529

Numbers in parentheses are standard errors. *** ** and * indicate significance at the 1%, 5% and 10% levels, respectively. The state-owned corporations are Swedish Railways, Swedish Maritime Administration, Swedish Civil Aviation Administration and Sweden's power-supply contingency planning authority.

Human capital is found to be the most important predictor of ethical screening. This resembles findings on the supply of volunteer labor (Freeman, 1997) and charity contributions (Andreoni, 2008). Unlike these activities, however, ethical screening is not influenced by wealth or income. One important difference is the degree of anonymity. As the act of applying an ethical screen is not easily visible to others, screening appears not to be the result of prestige-seeking.

Shleifer (2004) argues that competition is the underlying force behind the spread of unethical behavior and suggests that, as competition increases income, more will pay for ethical behavior. The results presented here suggest that ethical screening is not sensitive to income, but rather to education. Competition may therefore only indirectly increase ethical behavior if higher income leads to more education.

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References

- Andreoni, J., 2008. Charitable giving. In: Durlauf, S.N., Blume, L.E. (Eds.), *The New Palgrave Dictionary of Economics*, 2nd Edition. Palgrave Macmillan.
- Freeman, R.B., 1997. Working for nothing: the supply of volunteer labor. *J. Labor Econ* 15 (1), 140–166.
- Shleifer, A., 2004. Does competition destroy ethical behavior? *American Economic Review* 94 (2), 414–418.
- Social Investment Forum, 2007. *The 2007 Report on Socially Responsible Investing Trends in the United States*. Social Investment Forum, Washington DC.
- Titmuss, R.M., 1970. *The Gift Relationship from Human Blood to Social Policy*. New Press, London.

³ One explanation could be mere occupational self-interest. However, a two-sided t -test to assess the difference in the share of medical funds chosen by health workers against other ethical investors is negative and statistically significant.

⁴ However, a two-sided t -test shows that the small minority who screen their entire portfolio are, on average, older, wealthier and more likely to work in the health sector compared to other ethical investors.